FitTrack AI — Project Design & Architecture

Phase 3 Technical Documentation

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Architecture Pattern: Modular Model-View-Controller (MVC)

Core Stack: Node.js, Express, MongoDB, Gemini 2.5 Flash API

1. Executive Summary & Overview

FitTrack AI is an intelligent fitness tracking platform designed to deliver personalized health insights, fitness activity logging, and community interactions. The backend architecture follows a modular MVC design, offering scalability, clear separation of concerns, and robust security through JWT authentication and password hashing.

2. System Architecture & Component Design

The system is structured around three primary layers:

- **Controllers Layer:** Handles HTTP requests, input validation, and maps API endpoints to core logic.
- **Services Layer:** Encapsulates business logic, rule-based fallbacks, and integration with the Gemini 2.5 Flash API.
- **Data Access Layer (Models):** Defines Mongoose schemas for MongoDB, handling user profiles, activity records, and goal tracking.

> **Key Architectural Highlight:** A fallback mechanism ensures continuous service. If external AI API services are unresponsive, the system gracefully degrades to rule-based workout and nutrition advice without breaking user workflows.

3. Data Models & Database Schema

Model	Primary Fields	Description
User	`_id`, `name`, `email`, `passwordHash`, `role`	Stores user credentials, profile metadata, and security tokens.
Activity	`_id`, `userId`, `type`, `duration`, `calories`, `timestamp`	Logs workouts, duration, metrics, and energy expenditure.
Recommendation	`_id`, `userId`, `insightText`, `source`	

Model	Primary Fields	Description
		Stores generated AI fitness advice and fallback rules.

4. Security & API Endpoints

All sensitive routes require a Bearer JWT token in the authorization header. Password storage employs `bcrypt` for salt hashing prior to database persistence.

- `POST /api/auth/register` — User onboarding and password hashing.
- `POST /api/auth/login` — User authentication yielding a JWT token.
- `GET /api/activities` — Retrieves logged workouts for the authenticated user.
- `POST /api/recommendations/generate` — Triggers AI-assisted or fallback workout analysis.

5. AI Integration Strategy

FitTrack AI leverages the **Gemini 2.5 Flash model** to analyze user workout trends and generate structured wellness recommendations. In cases of API throttling or connectivity loss, pre-configured rule engines calculate standard caloric expenditures and default fitness routines to maintain service uptime.